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UNIVERSITI TEKNOLOGI MALAYSIA
Faculty of Artificial Intelligence

TOPIC 9:
SUPERVISED LEARNING MODELLING AND EVALUATION: CLASSIFICATION USING NAÏVE BAYES

MACHINE LEARNING (SAIA 2113)

Session 2025/2026 - Sem 1

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Topic 9: Supervised Learning Modelling And Evaluation: Classification Using Naïve Bayes

- Introduction to Naïve Bayes Classification
- Application, Types of Naïve Bayes & Bayes' Theorem
- Advantages and Disadvantages of Naïve Bayes Classifier

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Introduction

- **Naive Bayes** is a **supervised machine learning algorithm** used mainly for **classification tasks**.

- It is based on Bayes' Theorem — a rule that calculates the probability of an event based on prior knowledge of conditions related to the event.

- Uses probability to classify data. Picks the class with the higher probability.

- "Naive": Because it naively assumes feature independence, even though, in reality, features might be related. (e.g., "win" and "prize" are treated as unrelated).

- "Bayes": Because it is based on Bayes' Theorem, developed by mathematician Thomas Bayes.

Introduction

	Outlook	Temperature	Humidity	Windy	Play Golf
0	Rainy	Hot	High	False	No
1	Rainy	Hot	High	True	No
2	Overcast	Hot	High	False	Yes
3	Sunny	Mild	High	False	Yes

The dataset is divided into two parts, namely, **feature matrix** and the **response vector**.

- **Feature matrix:** contains all the vectors(rows) of dataset in which each vector consists of the value of **dependent features**. In above dataset, features are 'Outlook', 'Temperature', 'Humidity' and 'Windy'.
- **Response vector:** contains the value of **class variable** (prediction or output) for each row of feature matrix. In above dataset, the class variable name is 'Play golf'.

Assumption

- **Feature independence:** This means that when we are trying to classify something, we assume that each feature (or piece of information) in the data does not affect any other feature.
- **Continuous features are normally distributed:** If a feature is continuous, then it is assumed to be normally distributed within each class.
- **Discrete features have multinomial distributions:** If a feature is discrete, then it is assumed to have a multinomial distribution within each class.
- **Features are equally important:** All features are assumed to contribute equally to the prediction of the class label.
- **No missing data:** The data should not contain any missing values.

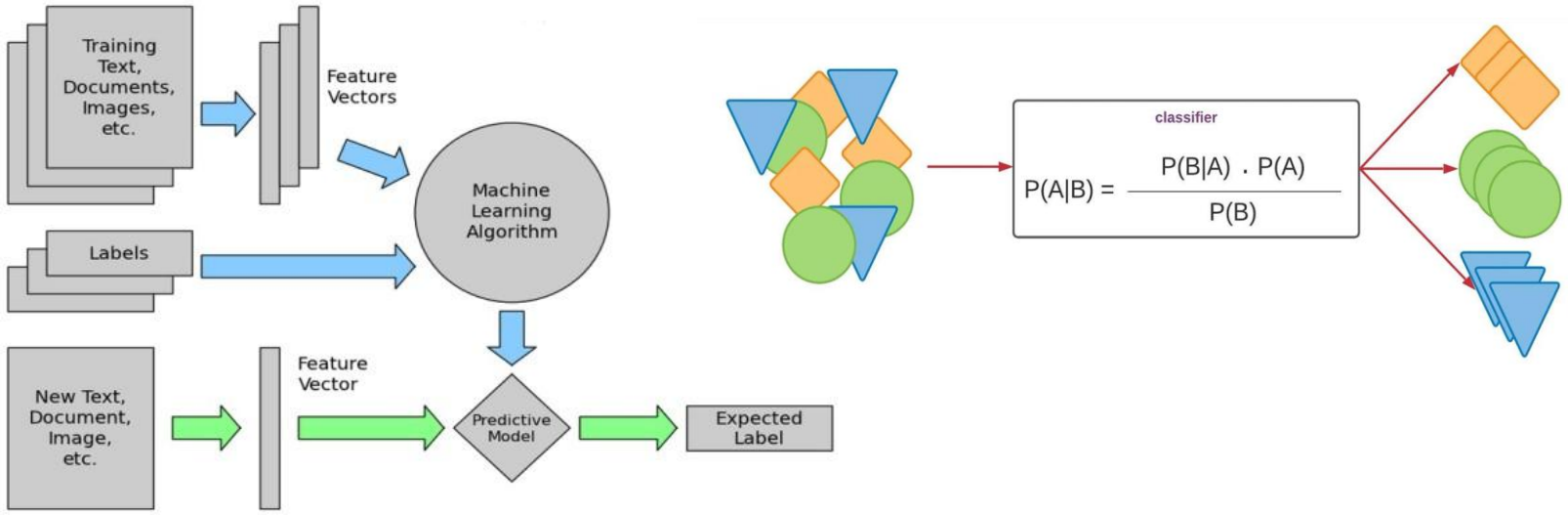
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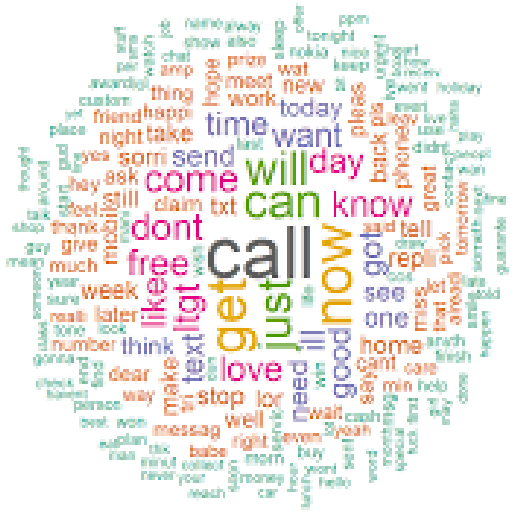
Application of Naïve Bayes Classification

- **Text Classification**
- **Spam Filtering**
- **Hybrid Recommender System**
- **Online Application**



Application of Naïve Bayes Classification

- **Text Classification**
- **Spam Filtering**
- **Hybrid Recommender System**
- **Online Application**



- Used to classify emails as spam or ham (non-spam) based on word frequency.
- Words like “free,” “win,” or “prize” might increase the probability of spam.

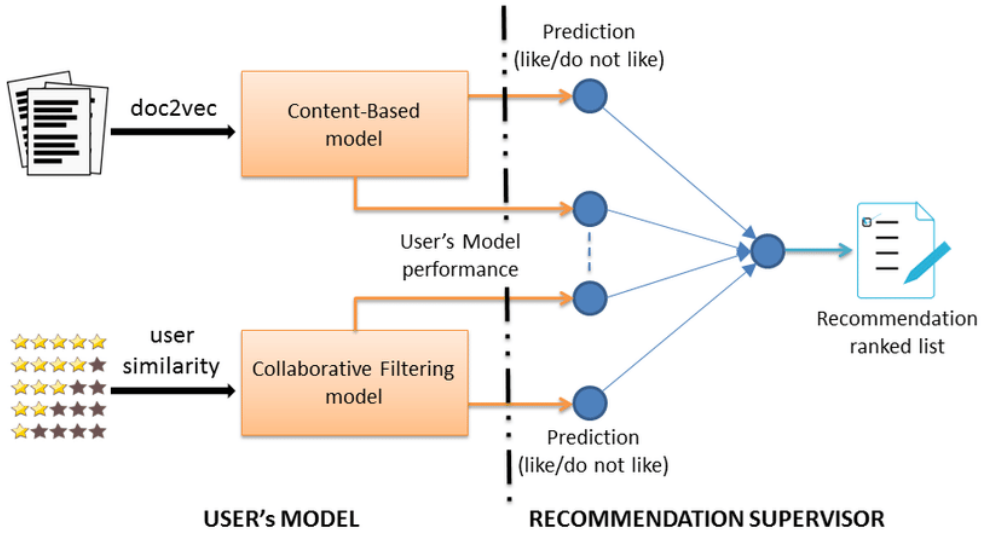
Question:

What is the example of words mostly use in spam email?

Application of Naïve Bayes Classification

- **Text Classification**
- **Spam Filtering**
- **Hybrid Recommender System**
- **Online Application**

- Recommender Systems apply machine learning and data mining techniques for filtering unseen information and can predict whether a user would like a given resource.
- Spotify using both user listening history and popular songs among similar users.
- Platforms like Netflix or Amazon can use this algorithm to suggest movies, TV shows, o heir previous behaviour

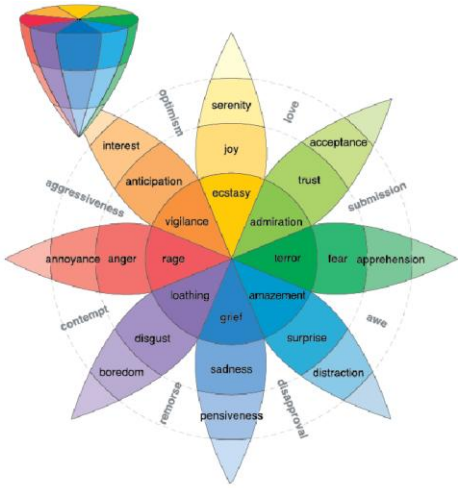


Application of Naïve Bayes Classification

- **Text Classification**
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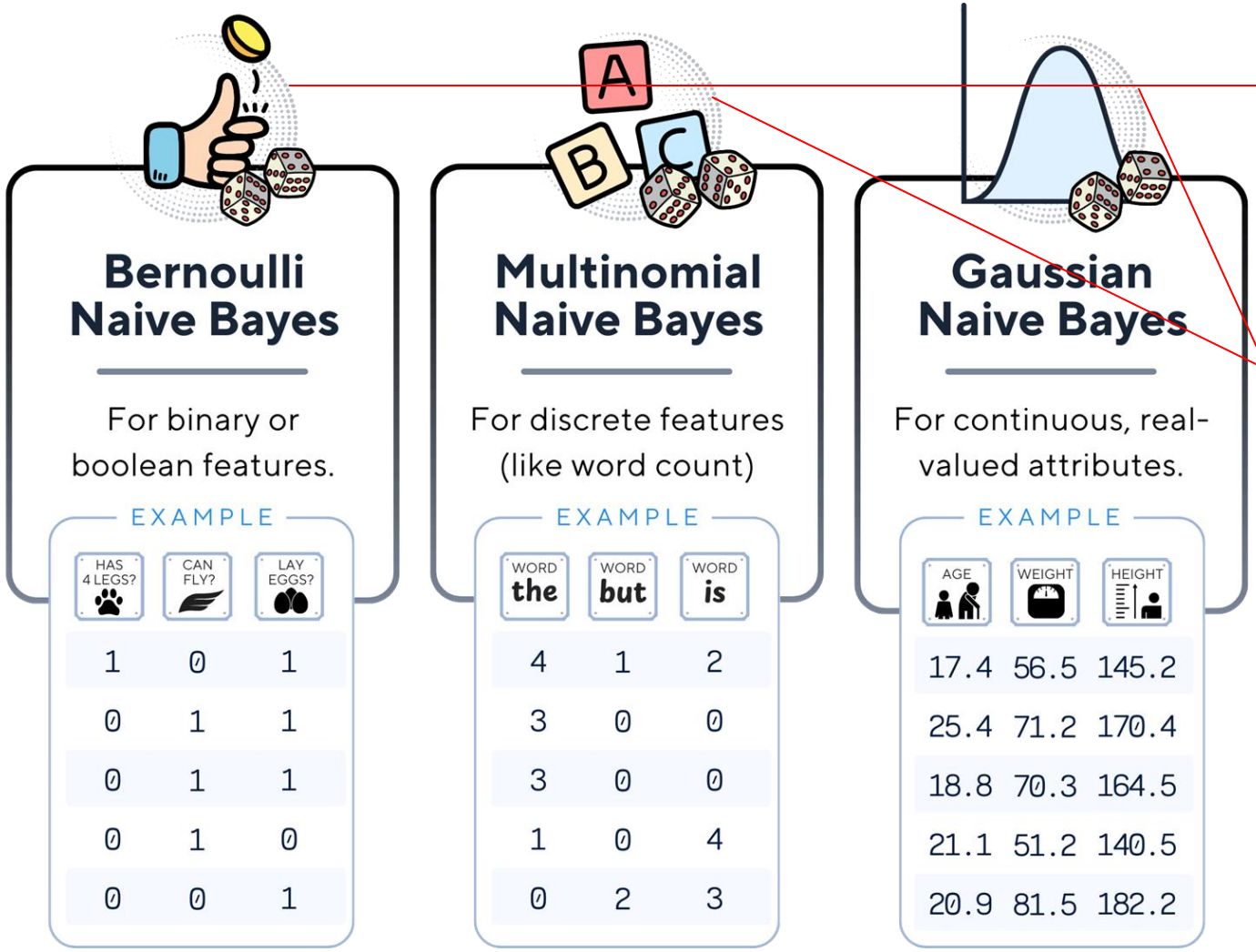
(a) 6 basic emotions



(b) Componential models

- Simple emotion modelling
- Emotion modeling refers to the process of detecting, analyzing, and responding to human emotions in digital applications.
- Chatbots, customer service, mental health apps, gaming, and e-learning to enhance user experience.

Types of Naïve Bayes Model



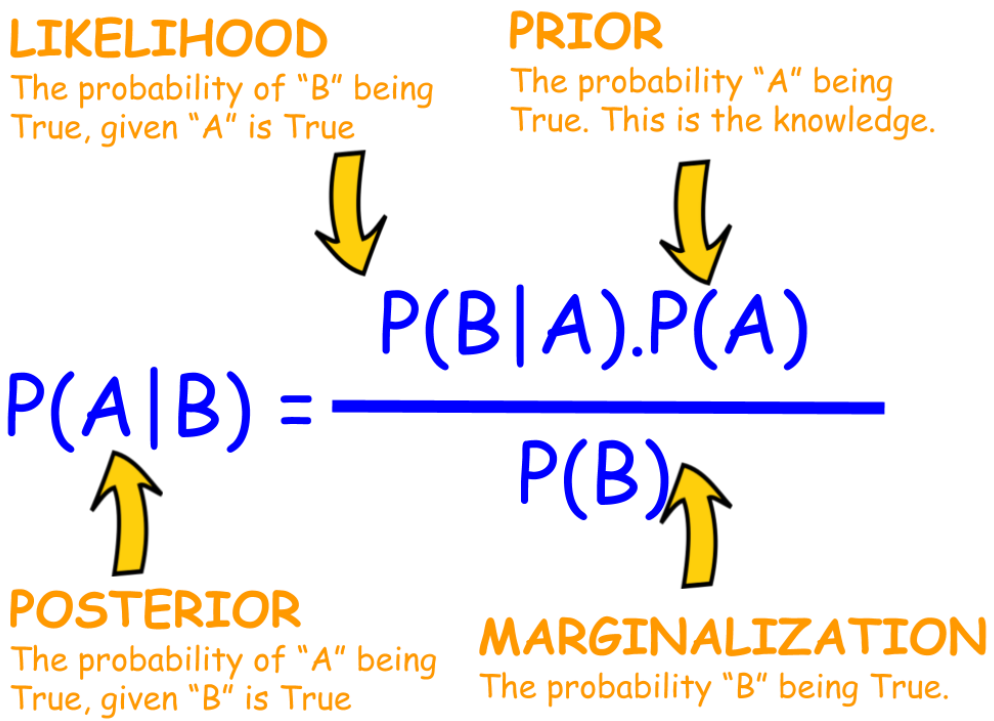
- Deals with binary features.
- Used with Boolean variables, that is, variables with two values, such as True and False or 1 and 0.

- Assumes that the features are from multinomial distributions.
- Useful when using discrete data, such as frequency counts
- Such as word counts in a document, natural language processing use cases, like spam classification.

- Used with Gaussian Distributions (i.e. normal distributions and continuous variables).
- Fitted by finding the mean and standard deviation of each class.
- Bell-shaped curve which is symmetric about the mean of the feature values

Bayes' Theorem

- Bayes' theorem is also known as **Bayes' Rule** or **Bayes' law**, which is used to determine the probability of a hypothesis with prior knowledge. It depends on the conditional probability.
- The formula for Bayes' theorem is given as:



Where:

- **P(A|B) is Posterior probability**: Probability of hypothesis A on the observed event B.
- **P(B|A) is Likelihood probability**: Probability of the evidence given that the probability of a hypothesis is true.
- **P(A) is Prior Probability**: Probability of hypothesis before observing the evidence.
- **P(B) is Marginal Probability**: Probability of Evidence.

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

Example 1

Given an example of weather conditions and playing sports. Calculate the probability of playing sports.

Now, you need to classify whether players will play or not, based on the weather condition.

First Approach (In case of a single feature)

Naive Bayes classifier calculates the probability of an event in the following steps:

- **Step 1**: Calculate the prior probability for given class labels
- **Step 2**: Find Likelihood probability with each attribute for each class
- **Step 3**: Put this value in Bayes Formula and calculate posterior probability.
- **Step 4**: See which class has a higher probability, given the input belongs to the higher probability class.

For simplifying prior and posterior probability calculation, you can use the two tables frequency and likelihood tables. Both of these tables will help you to calculate the prior and posterior probability.

The Frequency table contains the occurrence of labels for all features. There are two likelihood tables.

Likelihood Table 1 is showing prior probabilities of labels and Likelihood Table 2 is showing the posterior probability.

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Example 1

	Weather	Play
1	Sunny	No
2	Sunny	No
3	Overcast	Yes
4	Rainy	Yes
5	Rainy	Yes
6	Rainy	No
7	Overcast	Yes
8	Sunny	Yes
9	Sunny	Yes
10	Rainy	No
11	Sunny	Yes
12	Overcast	Yes
13	Overcast	Yes
14	Rainy	No

Frequency Table

Whether	No	Yes
Overcast		4
Sunny	2	3
Rainy	3	2
Total	5	9

Likelihood Table 1

Whether	No	Yes		
Overcast		4	=4/14	0.29
Sunny	2	3	=5/14	0.36
Rainy	3	2	=5/14	0.36
Total	5	9		
	=5/14	=9/14		
	0.36	0.64		

Likelihood Table 2

Whether	No	Yes	Posterior Probability for No	Posterior Probability for Yes
Overcast		4	0/5=0	4/9=0.44
Sunny	2	3	2/5=0.4	3/9=0.33
Rainy	3	2	3/5=0.6	2/9=0.22
Total	5	9		

Example 1

Now suppose you want to calculate the probability of playing when the weather is overcast.

Probability of playing: $$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

P(Yes | Overcast) = *P(Overcast | Yes)* P(Yes) / P (Overcast) .....(1)

Calculate Prior Probabilities:

a. P(Overcast) = 4/14 = 0.29
b. P(Yes)= 9/14 = 0.64

Calculate Posterior Probabilities:

a. P(Overcast |Yes) = 4/9 = 0.44

Put Prior and Posterior probabilities in equation (1)

a. P (Yes | Overcast) = 0.44 * 0.64 / 0.29 = 0.98(Higher)

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Example 1

Probability of not playing:

a. *P(No | Overcast) = P(Overcast | No)* P(No) / P (Overcast) .....(2)

Calculate Prior Probabilities:

a. P(Overcast) = 4/14 = 0.29
b. P(No)= 5/14 = 0.36

Calculate Posterior Probabilities:

a. P(Overcast |No) = 0/5 = 0

Put Prior and Posterior probabilities in equation (2)

a. P (No | Overcast) = 0 * 0.36 / 0.29 = 0
b. *The probability of a 'Yes' class is higher. So you can determine here if the weather is overcast than players will play the sport.*

Example 1

Second Approach (In case of multiple features)

Naive Bayes classifier calculates the probability of an event in the following steps:

- **Step 1**: Calculate the prior probability for given class labels
- **Step 2**: Calculate conditional probability with each attribute for each class.
- **Step 3**: Multiply same class conditional probability.
- **Step 4**: Multiply prior probability with step 3 probability.
- **Step 5**: See which class has higher probability, higher probability class belongs to given input set step.

	Weather	Temperature	Play
1	Sunny	Hot	No
2	Sunny	Hot	No
3	Overcast	Hot	Yes
4	Rainy	Mild	Yes
5	Rainy	Cool	Yes
6	Rainy	Cool	No
7	Overcast	Cool	Yes
8	Sunny	Mild	Yes
9	Sunny	Cool	Yes
10	Rainy	Mild	No
11	Sunny	Mild	Yes
12	Overcast	Mild	Yes
13	Overcast	Hot	Yes
14	Rainy	Mild	No

Example 1

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

Probability of playing:

*P(Play= Yes | Weather=Overcast, Temp=Mild) = P(Weather=Overcast, Temp=Mild | Play= Yes)*P(Play=Yes) .....(1)

P(Weather=Overcast, Temp=Mild | Play= Yes)= P(Overcast |Yes) P(Mild |Yes) ……..(2)

1. Calculate Prior Probabilities:

   P(Yes) = 9/14 = 0.64

1. Calculate Posterior Probabilities:

   P(Overcast |Yes) = 4/9 = 0.44

   P(Mild |Yes) = 4/9 = 0.44

1. Put Posterior probabilities in equation (2) P(Weather=Overcast, Temp=Mild | Play= Yes)

   = 0.44 * 0.44 = 0.1936(Higher)

1. Put Prior and Posterior probabilities in equation (1) P(Play= Yes | Weather=Overcast, Temp=Mild)

   = 0.1936*0.64 = 0.124

Example 1

Probability of not playing:

*P(Play= No | Weather=Overcast, Temp=Mild) = P(Weather=Overcast, Temp=Mild | Play= No)*P(Play=No) .....(3)

P(Weather=Overcast, Temp=Mild | Play= No)= P(Weather=Overcast |Play=No) P(Temp=Mild | Play=No) …….(4)

1. Calculate Prior Probabilities: P(No)= 5/14 = 0.36
2. Calculate Posterior Probabilities:

   P(Weather=Overcast |Play=No) = 0/9 = 0

   P(Temp=Mild | Play=No)=2/5=0.4

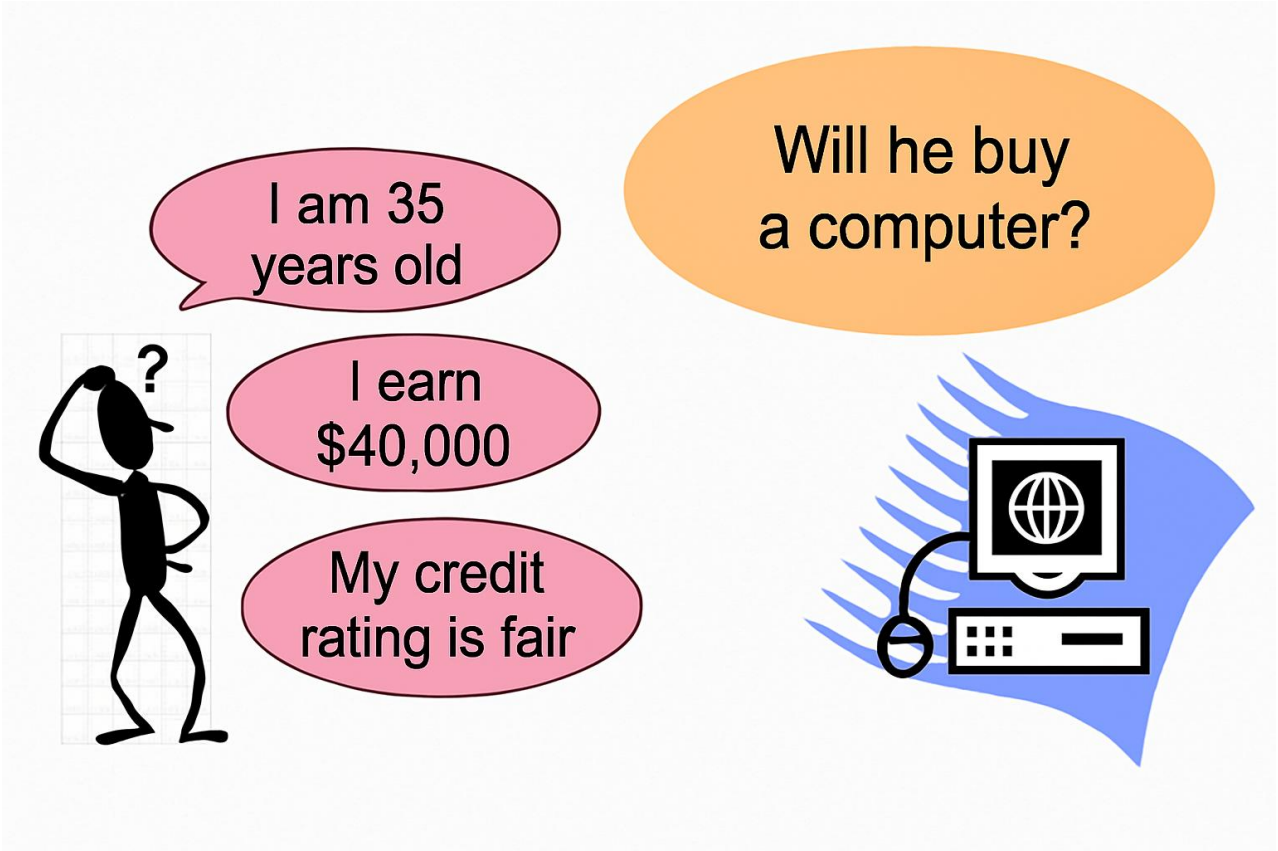
1. Put posterior probabilities in equation (4) P(Weather=Overcast, Temp=Mild | Play= No) = 0 * 0.4= 0
2. Put prior and posterior probabilities in equation (3) P(Play= No | Weather=Overcast, Temp=Mild) = 0*0.36=0

The probability of a 'Yes' class is higher. So you can say here that if the weather is overcast then players will play the sport.

Example 2

X = 35 years old customer with an income of $40,000 and fair credit rating.

H = Hypothesis that the customer will buy a computer.



Example 2

- The Bayes Theorem:

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

- P(A|B) : Probability that the customer will buy a computer given that we know his age, credit rating and income. (Posterior Probability of A)

- P(A) : Probability that the customer will buy a computer regardless of age, credit rating, income (Prior Probability of A)

- P(B|A) : Probability that the customer is 35 yrs old, have fair credit rating and earns $40,000, given that he has bought our computer (Posterior Probability of B)

- P(B) : Probability that a person from our set of customers is 35 yrs old, have fair credit rating and earns $40,000. (Prior Probability of B)

Example 2

Example: Class-Labeled Training Tuples from the *Customer Database*

RID	age	income	student	credit_rating	Class: buys_computer
1	youth	high	no	fair	no
2	youth	high	no	excellent	no
3	middle_aged	high	no	fair	yes
4	senior	medium	no	fair	yes
5	senior	low	yes	fair	yes
6	senior	low	yes	excellent	no
7	middle_aged	low	yes	excellent	yes
8	youth	medium	no	fair	no
9	youth	low	yes	fair	yes
10	senior	medium	yes	fair	yes
11	youth	medium	yes	excellent	yes
12	middle_aged	medium	no	excellent	yes
13	middle_aged	high	yes	fair	yes
14	senior	medium	no	excellent	no

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Example 2

X = (Age = '<=30', Income = medium, Student = yes, Credit_rating = fair)

- P(C1) = P(Buys_computer = yes) = 9/14 = 0.643
- P(C2) = P(Buys_computer = no) = 5/14 = 0.357
- P(Age='<=30' | Buys_computer = yes) = $\frac{\text{number of tuples with Buys_computer = yes and Age <=30}}{\text{number of tuples with Buys_computer = yes}}$
- P(Age='<=30' | Buys_computer = yes) = 2/9 = 0.222

Similarly,

- P(Age='<=30' | Buys_computer = no) = 3/5 = 0.600
- P(Income=medium | Buys_computer = yes) = 4/9 = 0.444
- P(Income=medium | Buys_computer = no) = 2/5 = 0.400
- P(Student=yes | Buys_computer = yes) = 6/9 = 0.667
- P(Student=yes | Buys_computer = no) = 1/5 = 0.200
- P(Credit_rating=fair | Buys_computer = yes) = 6/9 = 0.667
- P(Credit_rating=fair | Buys_computer = no) = 2/5 = 0.400

RID	age	income	student	credit_rating	Class: buys_c
1	youth	high	no	fair	no
2	youth	high	no	excellent	no
3	middle_aged	high	no	fair	yes
4	senior	medium	no	fair	yes
5	senior	low	yes	fair	yes
6	senior	low	yes	excellent	no
7	middle_aged	low	yes	excellent	yes
8	youth	medium	no	fair	no
9	youth	low	yes	fair	yes
10	senior	medium	yes	fair	yes
11	youth	medium	yes	excellent	yes
12	middle_aged	medium	no	excellent	yes
13	middle_aged	high	yes	fair	yes
14	senior	medium	no	excellent	no

Example 2

▶ P(X | Buys a computer = yes)

= P(Age='<=30' | buys_computer = yes) × P(Income=medium | buys_computer = yes) × P(Student=yes | buys_computer = yes) x P(Credit rating=fair | buys_computer = yes)

= 0.222 × 0.444 × 0.667 × 0.667 = 0.044

▶ P(X | Buys a computer = No)
= 0.600 × 0.400 × 0.200 × 0.400 = 0.019

▶ Find class C_i that Maximizes $P(X|C_i) \times P(C_i)$
→ P(X | Buys a computer = yes) × P(Buys_computer = yes) = 0.044 x 0.643 = 0.028
→ P(X | Buys a computer = No) × P(Buys_computer = no) = 0.019 x 0.357 = 0.007

▶ Prediction : Buys a computer for Tuple X

RID	age	income	student	credi
1	youth	high	no	fair
2	youth	high	no	excel
3	middle_aged	high	no	fair
4	senior	medium	no	fair
5	senior	low	yes	fair
6	senior	low	yes	excel
7	middle_aged	low	yes	excel
8	youth	medium	no	fair
9	youth	low	yes	fair
10	senior	medium	yes	fair
11	youth	medium	yes	excel
12	middle_aged	medium	no	excel
13	middle_aged	high	yes	fair
14	senior	medium	no	excel

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Advantages

1. Simple and Easy to Implement

- Requires relatively short training time compared to more complex algorithms.

2. Computationally Efficient

- Has a very low computation cost and requires fewer computational resources.
- Scales well with data, even for large datasets.
- Has low storage requirements.

3. Performs Well with Limited Training Data

- Effective even with small amounts of training data.
- Requires less data for training compared to other models like logistic regression or decision trees.

Advantages

4. Handles High-Dimensional Data

- Performs well with a large number of features.
- Particularly effective for use cases like document classification, where data is high-dimensional.

5. Strong Performance Despite Independence Assumption

- Despite assuming feature independence, it often delivers surprisingly accurate results.
- When the independence assumption holds, it can outperform models like logistic regression and decision trees.

6. Works Well with Categorical Features

- Particularly effective when the input features are categorical.
- Performs better with discrete variables compared to continuous variables.

Advantages

7. Can Handle Numerical Features

- Can work with numerical features by assuming they follow a normal (Gaussian) distribution — although this is a strong assumption.

8. Versatility

- Can be used for both binary and multi-class prediction problems.
- Successfully applied across various fields, especially in text analytics and document classification.

9. Ease of Interpretation

- Provides clear insight into how each feature contributes to the prediction through conditional probabilities.

Disadvantages

1. Strong Independence Assumption

- Assumes that all features are independent, which is often unrealistic in real-world data.
- In practice, predictors are often correlated (e.g., the presence of "snow" and "cold" are related in text data), violating the independence assumption and possibly leading to incorrect classifications.

2. Zero Frequency Problem

- If a categorical variable value in the test set was not observed in the training data, the model assigns zero probability to the entire prediction.
- This is known as the "Zero Frequency" or "Zero Probability" problem.

Disadvantages

3. Sensitivity to Irrelevant Features

- Naïve Bayes can be affected by irrelevant features or noise in the data, potentially impacting its accuracy.

4. Poor Probability Estimates

- Naïve Bayes is considered a bad estimator of class probabilities.

5. Performance on Numerical Data

- Naïve Bayes tends to perform better on categorical data.
- For numerical features, it assumes a normal (Gaussian) distribution, which may not always be valid and could affect performance.

THANK YOU



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